

High-Level Design (HLD)

Designer Catalogue & Inspiration Platform (Pinterest-like)

1. Document Overview

Role: Project Manager & Product Designer

Purpose: This document describes the high-level technical and product architecture for a Pinterest-like platform where designers create catalogues and users discover, save, and order designs.

Audience: - Founders / Stakeholders - Backend & Frontend Developers - UI/UX Designers - Investors / Partners

2. Product Vision

To build a **visual discovery and catalogue platform** that enables designers to: - Showcase collections professionally - Reach high-intent clients - Convert inspiration into inquiries or orders

And enables users to: - Discover designs - Save inspiration - Contact or order directly from designers

Core Value Proposition:

Inspiration + Portfolio + Catalogue + Commerce

3. User Roles & Permissions

3.1 Guest Users

- Browse public feed
- View catalogues & designs
- Search designs

3.2 Registered Users (Clients)

- Create boards
- Save designs & catalogues
- Follow designers
- Message designers
- Place inquiries / orders

3.3 Designers

- Create designer profile
- Create catalogues (collections)
- Upload designs (images/videos)
- Set pricing or inquiry-only
- View analytics
- Run promotions

3.4 Admin

- Verify designers
 - Moderate content
 - Feature catalogues
 - Manage subscriptions & ads
 - View platform analytics
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4. Functional Requirements

4.1 Feed & Discovery

- Pinterest-style masonry feed
- Trending designs
- Featured catalogues
- Personalized feed (phase 2)

4.2 Catalogue Management

- Create/Edit/Delete catalogues
- Catalogue cover image
- Multiple designs per catalogue
- Shareable catalogue links

4.3 Design Management

- Upload images/videos
- Tagging (style, occasion, fabric)
- Price / contact option
- Save & share design

4.4 Boards (Collections)

- Create boards
- Add designs or catalogues
- Public / Private visibility

4.5 Search & Filtering

- Keyword search
- Filter by:
 - Style
 - Category
 - Price
 - Designer

4.6 Messaging & Orders

- Client ↔ Designer chat
- Order request forms
- Status tracking

4.7 Monetization

- Designer subscription plans
 - Promoted catalogues
 - Transaction fees (optional)
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5. Non-Functional Requirements

- High availability
 - Scalable media storage
 - Fast feed loading
 - Secure authentication
 - Content moderation
 - Mobile-first UX
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6. System Architecture (High Level)

6.1 Architecture Pattern

Monolithic Backend (Phase 1) - Django + Django REST Framework

Microservices (Phase 2 – Optional) - Feed service - Messaging service - Payment service

7. High-Level System Components

7.1 Client Layer

- Web App (Django Templates / Next.js)

- Mobile App (Future – React Native / Flutter)

7.2 API Layer

- REST APIs
- JWT Authentication
- Role-based access control

7.3 Application Layer

- User Management
- Designer Catalogue Engine
- Feed & Recommendation Engine
- Messaging Service
- Subscription & Billing

7.4 Data Layer

- PostgreSQL (primary database)
 - Redis (caching & feed ranking)
 - Object Storage (Cloudinary / S3)
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8. Database Design (High-Level Entities)

Core Entities

- User
- DesignerProfile
- Catalogue
- Design
- Board
- SavedItem
- Message
- Subscription
- AdCampaign

Key Relationships

- DesignerProfile → User (One-to-One)
 - Catalogue → DesignerProfile (Many-to-One)
 - Design → Catalogue (Many-to-One)
 - Board → User (Many-to-One)
 - SavedItem → User → Design/Catalogue
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9. Feed Generation (HLD)

Phase 1 (Rule-Based)

- Latest designs
- Most saved designs
- Featured catalogues
- Followed designers

Phase 2 (Smart Ranking)

- User interaction history
 - Tag similarity
 - Engagement scoring
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10. Security & Access Control

- JWT-based authentication
 - Role-based permissions
 - Media upload validation
 - Rate limiting
 - Admin moderation tools
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11. Scalability Strategy

- CDN for media delivery
 - Redis caching
 - Asynchronous tasks (Celery)
 - Horizontal scaling on cloud
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12. Deployment Architecture

- Backend: Django (Dockerized)
 - Database: Managed PostgreSQL
 - Media: Cloudinary / AWS S3
 - Hosting: Render / AWS / Vercel (frontend)
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13. Analytics & Monitoring

- User engagement metrics
- Catalogue performance

- Conversion tracking
 - Error & performance monitoring
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14. MVP Scope

Included

- Designer profiles
- Catalogues
- Feed
- Boards
- Messaging

Excluded (Phase 2)

- In-app payments
 - AI recommendations
 - Mobile apps
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15. Risks & Mitigation

Risk	Mitigation
Heavy media usage	CDN & compression
Low designer adoption	Free tier & onboarding
Feed performance	Caching & pagination

16. Future Enhancements

- AI-powered recommendations
 - In-app checkout
 - Creator monetization
 - Brand collaborations
 - Mobile applications
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End of HLD Document